Version Control

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# Ysbyty Gwynedd Anaesthetics Department

# Entonox: Ward Administration

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**Purpose of Issue/Description of current changes:**

- To update best practice
- Added treatment regime for nitrous oxide toxicity
- Added Appendix 2. – Patient Assessment for use of Entonox

**<u>Purpose of issue/description of current changes:</u>**

- To update best practice
- Facemask first line, mouthpiece only if facemask not tolerated
- Environmental implications

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# 1. Introduction

Entonox (nitrous oxide) is a gaseous mixture of 50% nitrous oxide ($N_20$) and 50% oxygen (O2). It is patient controlled, inhaled analgesia which is used for the short-term relief of acute pain (Young et al., 2012). Nitrous oxide is a colourless, sweet-smelling gas with powerful analgesic properties, supplied in premixed cylinders (Brown & Sneyd, 2015). The gas is inhaled and self administered by the patient using a demand valve system attached to a mouthpiece (Collins, 2018). The nitrous oxide component of the gas acts as an analgesic, producing similar physiological effects to opioids (Bonafé-Monzó et al., 2015), whilst the oxygen component has an anti-hypoxic effect and ensures good cerebral perfusion and enhanced recovery (Peate & Lancaster, 2000). The technique is inherently safe as it is self-administered, and if the patient becomes too drowsy the mouthpiece will fall away from the patients face. Entonox® causes minimal respiratory depression (Schug et al., 2021).

Owing to its physiochemical properties, nitrous oxide has a rapid onset and short duration of action – some effects will be seen after four or five deep breaths. Offset of the effect is also rapid, so analgesic can only be maintained by repeated inhalations. It is therefore suitable for the provision of analgesia during labour and during painful procedures such as dressing changes and in the pre-hospital setting (Schug et al., 2021). It is often used in combination with opioid or other analgesics therapies. Due to environmental concerns the use of nitrous oxide has been limited in general wards.

A one-way valve allows delivery of the gas when the patient inspires, providing there is an airtight fit between the face and mouthpiece. The technique is inherently safe as it is self-administered, and if the patient becomes too drowsy the mouthpiece will fall away from the patients face. Entonox® causes minimal respiratory depression (Schug et al., 2021).

# 2. Aim

This document applies to all patients using Entonox in Betsi Cadwaladr University Health Board, and intended for all physicians, nurses, midwives, ODPs and pharmacists involved in the management of acute pain.

# 3. Objectives

- To manage pain effectively
- To protect and promote the safety and well-being of patients receiving Entonox
- To provide guidelines for the health practitioner undertaking the procedure
- To utilise evidence based safe practice

## 4. Responsibilities

General responsibilities of the multidisciplinary team:

1. To act in accordance with their own Professional Codes of Conduct
2. To follow agreed Health Board Drug Administration

## 5. Theory

The exact mechanism of action of nitrous oxide is unknown; its effects take place within the pain centres of the brain and spinal cord and related to the release of endogenous neurotransmitters such as opioid peptides and serotonin and the activation of certain opioid receptors (Fang, Guo, Davies & Maze, 1997, Gillman & Litchtigfeld, 1998). Pulmonary transfer of nitrous oxide is rapid, with onset of effect in seconds and full analgesic effect within 1-2 minutes. It is also rapidly eliminated from the blood, via the lungs, when inhalation ceases (Trojan *et al*., 1997).

## 6. Rationale

There are several advantages to using Entonox:

- As the gas is inhaled, it is painless when compared to systemic methods of administering analgesia
- Rapid onset of effect
- Few side effects and self-limiting as the gas is self-administered
- Does not depress respiratory or cardiovascular function when used as directed
- Effects wear off rapidly, usually within 5 minutes
- The patient is in full control providing reassurance that they have instant self-regulated access to analgesia, which also provides a focus of distraction from the procedure. They can both help reduce anxiety. Entonox also has sedative properties which can act as anxiolytic (BOC, 2019).

Because of these properties, Entonox is only for short-term pain relief following injury or trauma and during therapeutic and investigative procedures. Although its use it mainly in emergency care

## 7. Indications

- Wound and burn dressings, wound debridement.
- Changing and removing packs and drains
- Removal of sutures from sensitive areas.
- Invasive procedures e.g. cauterisation and sigmoidoscopy
- Removal of radioactive intra-cavity gynaecological applicators
- Altering the position of a patient who experience incident pain
- Physiotherapy procedures, particularly post-operatively
- Manual evacuation of the bowel in severe constipation
- For short-term use only i.e. no longer than 24 hours (NICE, 2023)

## 8. Contraindications

- Maxillofacial injuries due to surgery or local disease-altering anatomy (BOC, 2015)
- Heavily sedated patients, as they would be unable to breathe in the Entonox on demand, and to potentiate sedation further may be hazardous
- Intoxicated patients or those with impaired level of consciousness related to either to disease state or concurrent medications. Aspiration would be a hazard in the event of vomiting.
- Within 24 hours of laparoscopic surgery
- Allergy to nitrous oxide
- Any condition in which gas is entrapped within the body and where its expansion would be dangerous. such as:
  - Pneumothorax (artificial, traumatic or spontaneous)
  - Air embolism
  - Severe bullous emphysema
  - Abdominal distension or bowel obstruction (Schug et al., 2021)
  - Decompression sickness or following recent dive
  - Following air encephalography
  - During myringoplasty
  - Laryngectomy patients, as unable to use apparatus
  - In patients who have received recent intraocular injection of gas (BOC, 2019).
- Temperatures of below -6 °C as the gases separate (BOC, 2016).
- Not for prolonged use

- The nitrous oxide constituent of Entonox passes into any air-filled cavity within the body faster than nitrogen passes out. As air expands, this is likely to result in a build-up of tension, which will increase the patient's symptoms.

## 9. Use during pregnancy and lactation

- It is suggested that prolonged exposure to high levels of nitrous oxide may affect a women's ability to become pregnant (Axelsson et al., 1996). However, there is no published material to show that nitrous oxide to the human foetus, therefore there is no absolute contraindication to its use in the first 16 weeks of pregnancy.
- May depress neonatal respiration if used during delivery (NICE, 2023)
- There are no known adverse effects to using Entonox during breast-feeding (BOC, 2019).

## 10. Cautions

Prior to using Entonox, consideration should be given to the following:

- The high levels of oxygen (50%) of Entonox may depress respiration in patients who have chronic obstructive pulmonary disease and are carbon dioxide retainers
- Patients with poor intake or a diet low in animals' products e.g. vegan
- Patients with malabsorption syndromes, particularly those with ileal resections
- Patients requiring synthetic diets (egg. phenylketonuria, maple syrup syndrome)
- Patients on a diet with special vitamin or mineral supplements are prescribed for deficiency
- Entonox should not be used as a replacement for intravenous analgesia or general anaesthesia in procedures requiring increased levels of medical intervention.
- Additional care is needed when Entonox is administered to a patient who been given concomitant medication.
- If Entonox is used as the sole analgesic/sedative agent, it is recommended not to drive or operate complex machinery until
    - A healthcare professional judges the patient has returned to normal mental status
    - At least 30 minutes has elapsed after the administration of Entonox has ceased. (BOC, 2016).
- Entonox is highly addictive and should not be used for longer than 24-hours (NICE, 2023)

## 11. Professional Issues and Training

- To maintain professional safety, healthcare professionals need to understand the properties, applications and prescribing practices for medical product, which includes the prescribing practices for any medical product, including medical gases (BOC, 2011). Training should include:
    - Physical properties of the medical gas Entonox
    - Indications for its use
    - Side-effects and contraindications
    - Methods of safe administration of the gas

## 12. Prescribing

- Entonox can be administered without a written prescription from a Doctor providing there is a local Patient Group Directive (PGD) to allow this.
- If not, healthcare staff should seek medical advice and ensure Entonox is prescribed before giving it to a patient
- Non-medical prescribers who have completed the relevant training as an independent prescriber many also prescribe Entonox if it within their area of competence

## 13. Equipment

### Cylinders

- Entonox cylinders available from BOC in a variety of sizes
- All cylinders have blue and white markings on their shoulder.
- The large cylinders require a trolley to move them around the ward.
- The lightweight smaller cylinders are easier to carry, have a live content gauge, and easier to change because a regulator cylinder key is not necessary.
- Do NOT store next to oxygen cylinders to avoid confusion in an emergency situation

### Demand valves, filters, facemasks and mouthpieces

- Facemasks should be used first line, mouthpiece should only be used if a face mask cannot be tolerated
- Entonox equipment is a potential source of cross-infection; bacterial filters (single use only) should be fitted between the facemask or mouthpiece if the demand valve is not for single use (Chivers & Weisz, 2000).
- Facemasks and mouthpieces must also be single use and disposed of once therapy ceases
- Local policies must also be followed for the cleaning/sterilisation of non-disposable equipment between patients.



## 14. Assessment

The patient's ability to administer Entonox safely and effectively must be assessed prior to use. Patients must be able to:

- Understand the instructions for Entonox us
- Hold the demand valve to self-administer the gas
- Inhale the gas through the mouthpiece whilst breathing normally (patients with impaired lung function may not be able to inhale the gas sufficiently to provide adequate analgesia).

## 15. Pharmacological Support

- Entonox is designed for self-administration by the patient.
- This method of administration makes use of a demand unit which safeguards the patient from excessive inhalation of Entonox
- The demand unit ensures that gas can be obtained only be the patient inhaling from the mouthpiece and producing a negative pressure
- The gas flow stops when the patient stops inhaling and removes the mouthpiece.
- Thus patients are able to self-regulate the dose of Entonox (a method of patient-controlled analgesia).
- Therefore, the patient must hold the mask firmly over the lips to the mouthpiece to produce an airtight fit and breathe in before the gas will flow.
- Expired gases escape by the expiratory valve on the hand piece.
- It is essential to adhere to this method of self-administration as it is impossible for patients to overdose themselves because, if the become drowsy, they will relax on the handset and the gas flow will cease when no negative pressure is applied.
- However, should inhalation continue, light anaesthesia supervenes and the mouthpiece drops away as the patient relaxes.
- Entonox may be self-regulated, but additionally may be administered by attendant medical personal trained in its use e.g. obstetric/accident and emergency units.

## 16. Duration and frequency of administration

- The duration and frequency of Entonox should be tailored to the individual patients needs. Prolonged exposure to Entonox causes inactivation of levels and impaired folate metabolism and pernicious anaemia (BOC, 2019), it is recommended that:
- Entonox is used short-term, rather than on a long-term basis.
- It should not be used for more than 24 hours in total, or more frequently than once every 4 days (BOC, 2019).
- If daily use is required for more than 4 days, this should be closely accompanied by close supervision and blood monitoring to check for changes in red and white blood cells (BOC, 2019). Consideration should also be given to the administration of $B_{12}$ and folate supplements. If concerned, seek the advice of a Haematologist.

## 17. Complications

- Effects such as euphoria, disorientation, sedation, nausea, vomiting, dizziness and generalised tingling are commonly described. These are generally minor and rapidly reversible (BOC, 2019).

- Inappropriate inhalation of Entonox will ultimately result in unconsciousness, passing through stages of light headedness and intoxication. (BOC, 2019)
- Nitrous oxide oxidizes vitamin $B_{12}$ inactivating the enzyme methionine synthetase, essential for the synthesis of DNA and RNA (Sanders et al., 2008). This may lead to bone marrow depression (resulting in a megaloblastic anaemia) and a reduction in the synthesis of myelin (resulting in progressive myeloneuropathy). The clinical features resemble those of vitamin $B_{12}$ deficiency. However, vitamin $B_{12}$ levels may be normal, as the problem is due to a reduction in *active* vitamin $B_{12}$, levels and not necessarily total body levels. (Schug et al., 2021).

- Bone marrow toxicity is progressive but also reversible and almost completely preventable by the administration of folic acid (Oussalah et al, 2019). Neurotoxicity resulting from nitrous oxide use is rare, but can develop rapidly and may be irreversible. The clinical features are the same as those seen with vitamin B12deficiency-induced subacute combined degeneration of the spinal cord leading to numbness, paraesthesia, ataxia, and spasticity. The risk of this complication is significantly increased in patients with pre-existing vitamin B12 deficiency (e.g. some vegetarians or elderly), and the deficiency maybe subclinical, that is, detectable only by measuring vitamin $B_{12}$ blood levels. If concerned, seek the advice of the Haematologist.

- It was commonly thought that these complications may only follow prolonged or repeated administration of nitrous oxide. However, severe irreversible neurotoxicity has been reported after single, short-term exposure in susceptible patient (Schug et al., 2021).

## 18. Environmental Implications

Staff should limit their occupational exposure to nitrous oxide. Human factors are critical in reducing exposure, such as:

- Providing clear instructions to patients on correct use of equipment being used, including exhaling into the re-breather mask or out through the mouthpiece

- Staff positioning relative to exhaust $N_2O$ and the direction of ventilation flow
- Turning gas and air off when not in use
- Unplugging regulators from outlets when not in use
- Monitoring the condition of equipment for leakages.

# KEY POINTS FOR NITROUS OXIDE

1. Nitrous oxide is an effective analgesic for short-term use in painful procedures and in the pre-hospital setting. Nitrous oxide is also used to provide analgesia during labour.

2. Nitrous oxide is contraindicated in patients with air-containing spaces (e.g. pneumothorax, pneumocephalus, and bowel obstruction) or recent gas embolism.

3. Nitrous oxide can cause rare but serious complications related to myeloneuropathy bone marrow suppression. These complications are increased in patients with vitamin $B_{12}$ deficiency and may be reduced by limiting the time of exposure.

4. If a diagnosis of $N_20$ toxicity is suspected, treatment should be initiated before the return of definitive diagnostic results. A suggested regime is given below:

    - **Vitamin $B_{12}$ 1mg intramuscularly once daily**
    - **Folic once 5mg orally once daily (Royal College of Emergency Medicine, 2023)**

5. For more information on Nitrous Oxide (toxbase.org)

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# Appendix 1.

## Procedure for Entonox administration

| Essential Equipment | |
|---|---|
| • Entonox cylinder<br>• Sterile bacterial filter<br>• Mouthpiece<br>• Method of documentation | |
| **Pre-procedure** | |
| **Action** | **Rationale** |
| 1. Explain and discuss the procedure with the patient | To ensure that the patient understands the procedure and gives their valid consent (NMC, 2018)<br><br>To reduce patient anxiety (Royal College of Anaesthetics, 2006)<br><br>To ensure the patient has no underlying medical problems that contraindicate the use of Entonox (BOC, 2018) |
| 2. Turn on the Entonox supply from the cylinder | To ascertain whether there is any Entonox in the cylinder |
| 3. Examine the gauge to determine how much gas is in the cylinder | To ensure an adequate supply of gas throughout the procedure |
| 4. To ensure the patient is in as comfortable position as possible | To promote comfort |
| 5. Demonstrate how to hold the apparatus. by holding mouthpiece to you own face (then replace before handling patient equipment). Explain to the patient that when they breathe in and out regularly and deeply, a hissing sound will be heard, indicating the gas is being inhaled. | To ensure the patient understands what to do and what to expect |
| 6. Allow the patient to practice with the apparatus | To enable the patient to adopt the correct technique (BOC, 2013) |
| **Procedure** | |
| 7. Encourage the patient to breathe gas in and out for at least 2- 3 minutes, as this is the time it takes for the analgesic effect to take place (BOC, 2013). If the gas is being before a painful procedure, allow the patient to breathe the gas in for about 2 minutes before starting (BOC,2013) | To allow sufficient time for an adequate circulatory level of nitrous oxide to provide analgesia. When the patient inhales, gas first enters the lungs and then the pulmonary and systemic circulations. It takes 2-3 minutes to build up reasonable concentration of nitrous oxide in the brain |

| | |
|---|---|
| 8. During the procedure, encourage the patient to breath in and out regularly and deeply | To maintain adequate circulatory levels, thus providing adequate analgesia (BOC, 2015) |
| 9. Evaluate the effectiveness of Entonox with the patient throughout the procedure, by verbal questioning and encouraging the patient to self-assess the analgesic effect. This should them be documented. | To establish whether the Entonox has been a useful analgesic (NMC, 2018) |
| **Post-procedure** | |
| 10. Turn of the supply from the cylinder | To avoid potential leakage of gas from the equipment |
| 11. Depress the diaphragm under the demand valve | To remove residual gas from the tubing (BOC, 2013) |
| 12. Follow local guidelines for the cleaning of the demand valve and tubing. Filter and mouthpieces should be disguarded after use | To reduce the risk of cross-infection (BOC, 2011) |
| 13. At the end of the procedure, observe the patient for up to 30 minutes until the gases have worn off. | Some patients may feel transient drowsiness or giddiness and should be discouraged from getting out of bed until these effects have worn off (BOC, 2016, NMC,2018) |

# Appendix 2.




Addressograph

## Patient Assessment for the use of Entonox

Entonox is pain relief for short procedures only (drain removal, draining changes etc), and is not to be left unattended with the patient. To administer it requires a trained nurse and need to be prescribed on the 'once only' (stat) side of the patients prescription chart.

**Checklist:**

| | YES | NO | COMMENTS |
|---|---|---|---|
| Allergy to Entonox | | | |
| Laparoscopic surgery in last 24 hours | | | |
| Pneumothorax | | | |
| Air embolism | | | |
| Bullous Emphysema | | | |
| Gross abdominal distension/obstruction | | | |
| Intoxication | | | |
| Recent head injury/surgery/raised intracranial pressure | | | |
| Recent ear surgery/ infection | | | |
| Recent scuba dive | | | |
| Air encephalography | | | |

If the answer of any of the above is yes, the patient is not suitable for Entonox use and an alternative method should be sought. If the Entonox is needed more than once, please refer to pain management team. Regular use is discouraged as risk of haematological complication increased.

Signature: _____

Name: _____ Date: _____

**TO BE COMPLETED BY THE NURSE AND FILLED IN THE PATIENTS NOTES**